

COOLING WITHOUT POWER: ADVANCED PASSIVE DESIGN STRATEGIES FOR SUSTAINABLE HOME



A Project Report

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In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

MECHANICAL ENGINEERING

KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

(An Autonomous Institute)

ANNA UNIVERSITY :: CHENNAI-600 025

JAN 2026

BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

We thank our beloved Executive Chairman and President **Dr.PSS.SRINIVASAN, B.E., M.Tech., Ph.D.**, and all the members of Knowledge Institute of Technology Trust at this high time for providing us with excess of facilities to complete the project successfully.

My heartfelt gratitude to **Dr. K. VISAGAVEL, M.E., Ph.D., FIE**, Principal of KIOT, for his invaluable guidance and encouragement throughout the journey.

I am very much obliged to **Dr. K.S. PRABHAKARAN, M.E., Ph.D.**, Head of the Department, Mechanical Engineering, for his valuable insights that significantly enhanced the quality of my project.

I am humbled and thankful **Mr. S. SHANMUGAM, M.E.**, Professor of Practice, Department of Mechanical Engineering for Initiating the Institute To Industry (I2I) Course and Providing Valuable Industry Exposure . His guidance and motivation greatly enhanced our learning and contributed to the Successful completion of the project.

A sincere Thanks to my Project Work Supervisor, **Mr. A. KAMALAKKANNAN, M.E., (Ph.D).**, Department of Mechanical Engineering, Department of Mechanical Engineering, for his exceptional guidance and unwavering support throughout the project's duration.

I also wish to acknowledge the entire teaching faculty, supporting staffs of the Mechanical Engineering Department, and my fellow students for their assistance and cooperation, which played a vital role in bringing the project to fruition.

I am forever thankful to all who contributed for the successful completion of the project.

ABSTRACT

Rapid urbanization and increasing global temperatures have led to a higher dependence on mechanical cooling systems, resulting in excessive energy consumption and environmental impact. This project focuses on the design and development of a passive cooling home that maintains indoor thermal comfort without the use of electrical air-conditioning systems. The proposed structure utilizes a compact dome-shaped design with maximum dimensions of 600 mm × 600 mm × 600 mm, which enhances natural air circulation and minimizes heat gain. The system incorporates passive cooling techniques such as natural ventilation, thermal mass effect, controlled openings, and heat dissipation through structural geometry. Temperature sensors are used to monitor internal and external temperature variations, and the collected data is analyzed to evaluate cooling efficiency. The dome structure reduces direct solar heat absorption and promotes uniform airflow, resulting in a lower indoor temperature compared to the surrounding environment. This project demonstrates a low-cost, energy-efficient, and environmentally friendly housing solution, particularly suitable for hot and dry climatic regions. The results prove that passive cooling techniques can significantly reduce indoor temperatures and energy usage, making them a sustainable alternative for future residential designs. The results demonstrated that the indoor temperature remained consistently lower than the outdoor temperature, with a reduction of approximately 2–3°C during peak conditions. The study confirms that passive cooling methods can effectively improve thermal comfort while reducing energy consumption and environmental impact. The proposed system offers a low-cost, sustainable, and eco-friendly solution suitable for residential buildings in hot and tropical climates, providing an alternative to conventional air-conditioning systems. This study highlights the potential of passive cooling techniques as an alternative to conventional cooling systems and supports the development of sustainable and climate-responsive building designs suitable for hot and tropical regions.

Keywords: Passive Cooling, Natural Ventilation, Stack Effect, Venturi Effect, Dome Roof, Thermal Comfort, Sustainable Building, Energy Efficiency.

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LIST OF SYMBOLS, ABBREVLATIONS AND NOMENCLATURE

1. SYMBOLS:

SYMBOL	DESCRIPTION	UNIT
T_i	Indoor air temperature	$^{\circ}\text{C}$
T_o	Outdoor air temperature	$^{\circ}\text{C}$
ΔT	Temperature difference ($T_o - T_i$)	$^{\circ}\text{C}$
v	Air velocity	m/s
A	Cross-sectional area of airflow	m^2
Q	Volumetric airflow rate	m^3/s
ρ	Density of air	kg/m^3
C_p	Specific heat capacity of air	$\text{J}/\text{kg}\cdot\text{K}$
h	Convective heat transfer coefficient	$\text{W}/\text{m}^2\cdot\text{K}$
k	Thermal conductivity of material	$\text{W}/\text{m}\cdot\text{K}$
q	Heat transfer rate	W
I	Solar radiation intensity	W/m^2
ε	Emissivity of surface	—
α	Solar absorptivity	—
R	Thermal resistance	$\text{m}^2\cdot\text{K}/\text{W}$
ACH	Air changes per hour	h^{-1}

2. ABBREVIATIONS

ABBREVIATION	FULL FORM
PCM	Phase Change Material
AAC	Autoclaved Aerated Concrete
ACH	Air Changes per Hour
CFD	Computational Fluid Dynamics
HVAC	Heating, Ventilation and Air Conditioning
SHGC	Solar Heat Gain Coefficient
U-Value	Overall Heat Transfer Coefficient
TDS	Thermal Distribution System
RH	Relative Humidity
COP	Coefficient of Performance
LPD	Lighting Power Density

3. NOMENCLATURE

TERM	DEFINITION / DESCRIPTION
Passive Cooling	Cooling of indoor spaces using natural methods without electrical or mechanical systems
Dome Structure	Semi-cylindrical roof form used to enhance air circulation and reduce heat gain
Ventilation Opening	Designed inlet and outlet openings for natural airflow
Stack Effect	Natural air movement caused by temperature and density differences
Solar Reflective Coating	Surface coating that reflects solar radiation to reduce heat absorption
Indoor Temperature	Temperature measured inside the prototype or house
Outdoor Temperature	Ambient temperature surrounding the structure
Airflow Path	Direction followed by air entering and exiting the structure
Thermal Mass	Ability of materials to absorb and store heat
Insulation	Material used to reduce heat transfer through walls or roof
Heat Gain	Amount of heat entering the building from external sources
Heat Dissipation	Release of stored heat to the surroundings
Prototype Model	Small-scale cardboard model used for experimental analysis

CHAPTER 1

INTRODUCTION

Rapid urbanization, population growth, and rising global temperatures have significantly increased the demand for cooling in residential buildings. Conventional cooling systems such as air conditioners consume large amounts of electrical energy, leading to higher operating costs and increased environmental impact through greenhouse gas emissions. As energy consumption continues to rise, there is a growing need for sustainable and energy-efficient alternatives that can provide indoor thermal comfort while reducing dependence on mechanical cooling systems.

Passive cooling is an effective building design approach that utilizes natural processes such as ventilation, heat dissipation, thermal insulation, and solar heat control to maintain comfortable indoor temperatures without the use of electrical energy. Unlike active cooling systems, passive cooling techniques rely on architectural design, material selection, building orientation, and airflow management to reduce indoor heat gain and improve thermal comfort.

In tropical regions such as Tamil Nadu, buildings are exposed to intense solar radiation and high ambient temperatures throughout most of the year. Traditional building designs often fail to consider climate-responsive features, resulting in excessive heat accumulation and increased reliance on air-conditioning systems. Therefore, adopting passive cooling strategies has become essential for achieving sustainable and environmentally friendly residential construction.

This project presents the design and development of a passive cooling home prototype that aims to reduce indoor temperature through natural ventilation and heat rejection mechanisms. The proposed structure incorporates a semi-cylindrical dome-shaped roof, which promotes smooth airflow circulation, minimizes heat accumulation, and reduces direct solar heat gain. Additional passive cooling features such as stack-effect ventilation, Venturi-assisted airflow, reflective surface treatment, and strategically placed openings are integrated into the design to enhance cooling performance.

A scaled prototype was fabricated using lightweight materials such as cardboard and foam board to evaluate the effectiveness of the proposed cooling concept. Temperature sensors and data logging systems were incorporated to monitor indoor and outdoor temperature variations and assess the cooling performance of the structure under real environmental conditions. Experimental observations demonstrated that the indoor temperature remained lower than the surrounding outdoor temperature, confirming the effectiveness of the passive cooling approach.

The primary objective of this project is to develop a low-cost, energy-efficient, and environmentally sustainable cooling solution for residential buildings. By reducing dependence on mechanical cooling systems, the proposed passive cooling home contributes to energy conservation, lower carbon emissions, and improved thermal comfort. The findings of this study demonstrate the potential of passive cooling techniques as a practical and sustainable alternative for future housing developments in hot and tropical climates.

In recent years, the increasing demand for energy and the rapid growth of urban areas have resulted in a significant rise in electricity consumption across the world. One of the major contributors to this increase is the widespread use of air-conditioning systems in residential and commercial buildings. Although air conditioners provide thermal comfort, they consume a large amount of electrical energy and contribute to environmental issues such as carbon emissions and global warming. Therefore, there is a growing need to develop sustainable and energy-efficient cooling methods that can reduce energy consumption while maintaining comfortable indoor conditions.

Passive cooling is one of the most effective and environmentally friendly approaches for reducing indoor temperatures without the use of mechanical or electrical cooling equipment. Passive cooling techniques utilize natural phenomena such as airflow, heat transfer, thermal mass, insulation, shading, and building orientation to maintain thermal comfort. These techniques help reduce heat gain inside buildings and improve indoor air quality while minimizing operating costs and energy usage.

This project demonstrates that passive cooling techniques can provide a low-cost, sustainable, and energy-efficient alternative to conventional cooling systems. The proposed design can be further developed for full-scale residential applications using materials such as AAC blocks, reflective coatings, Low-E glass, and insulated doors. By reducing energy consumption and improving thermal comfort, passive cooling homes can contribute significantly to sustainable development and environmentally responsible construction practices

1.1 METHODOLOGY

This study adopts a systematic methodology to design, analyse, and evaluate a passive cooling house with the objective of reducing indoor thermal discomfort without the use of mechanical cooling systems. The methodology integrates architectural design principles, material selection, analytical heat transfer calculations, and computational fluid dynamics (CFD) simulations to assess the thermal and ventilation performance of the proposed structure.



Figure 1.1 Methodology

1.1.1 Climatic Data Collection and Analysis

The first step involves the collection of local climatic data for the study location (Salem, Tamil Nadu, India). Key parameters such as ambient temperature, solar radiation intensity, wind speed, wind direction, and relative humidity were obtained from standard meteorological databases. The climatic analysis helps in understanding seasonal thermal stress and wind availability, which directly influence passive cooling strategies.

1.1.2 Architectural Design of Passive Cooling House

Thermal conductivity and high thermal mass, such as AAC blocks and reflective surface coatings, were incorporated to reduce heat transfer into the indoor space. Roofing and wall materials were evaluated to limit solar heat gain during peak summer conditions.

1.1.3 Heat Transfer Analysis

Analytical heat transfer calculations were performed to estimate heat gain through the building envelope. The following modes of heat transfer were considered:

- **Conduction** through walls and roof
- **Convection** between indoor air and internal surfaces
- **Radiation** due to solar exposure

Steady-state heat transfer equations were used to calculate overall heat gain and expected indoor temperature reduction. These calculations provide a baseline for validating simulation results.

1.1.4 Natural Ventilation and Airflow Modelling

Natural ventilation performance was evaluated using airflow principles. Pressure differences generated due to wind-induced forces and temperature differences were analysed. The Venturi effect was considered at constricted openings to increase air velocity, thereby enhancing convective heat removal from the indoor environment.

Airflow rate was estimated using continuity and Bernoulli equations to determine velocity and pressure variations across inlet and outlet openings.

1.1.5 Computational Fluid Dynamics (CFD) Simulation

CFD simulations were conducted using ANSYS Workbench to analyse airflow patterns, temperature distribution, and velocity contours inside the passive cooling house. The building geometry was imported into the simulation environment, and air was defined as an incompressible fluid.

Boundary conditions included:

- Velocity inlet based on local wind data
- Pressure outlet at exhaust openings
- Heat flux or temperature boundary on walls and roof

The simulations enabled visualization of airflow circulation, thermal stratification, and cooling effectiveness within the indoor space.

1.1.6 Prototype Development and Experimental Validation

A scaled physical prototype of dimensions 60 cm × 60 cm was fabricated using lightweight materials to experimentally validate the proposed concept. Temperature sensors were placed at the inlet and outlet locations to measure indoor temperature variation under natural conditions. The experimental data were compared with analytical and CFD results to verify the reliability of the model.

1.1.7 Performance Evaluation and Comparison

The performance of the proposed passive cooling house was evaluated by comparing indoor temperature reduction, airflow velocity, and thermal comfort indices with those of a conventional house design. The results were analysed for different environmental conditions to assess the consistency and effectiveness of the passive cooling strategies.

1.1.8 Limitations and Assumptions

Certain assumptions were made to simplify the analysis, including steady-state conditions, uniform material properties, and neglect of internal heat gains from occupants and appliances. These limitations are acknowledged, and their potential impact on results is discussed.

1.1.9 Summary of Methodology

The overall methodology combines climate-responsive design, thermal analysis, numerical simulation, and experimental validation to comprehensively evaluate the passive cooling performance of the proposed house. This integrated approach ensures the technical credibility and practical applicability of the study.

CHAPTER 2

LITERATURE REVIEW

1. Hu et al. (2023) – Passive Design for Residential Buildings

Hu et al. conducted a systematic review of passive cooling strategies for residential buildings in hot climates. The study analysed 42 studies and found that passive design strategies could reduce indoor temperature by an average of **2.2°C**, cooling load by **31%**, and energy consumption by approximately **29%**. Roof, wall, window design and optimized cross-ventilation were identified as highly effective strategies. This study supports the use of an integrated passive cooling approach in the proposed home.

2. Kolani et al. (2023) – Passive Building Design in Tropical Climates

Kolani et al. reviewed research on passive building design and thermal comfort in tropical climates using bibliometric analysis. A total of 722 publications were analyzed to identify major research trends. The study highlighted natural ventilation, building design and thermal comfort as important areas for sustainable tropical buildings. These findings support the climate-responsive design approach used in the proposed passive cooling home.

3. Iskandar et al. (2024) – Natural Ventilation Using CFD

Iskandar et al. investigated different natural ventilation strategies in residential buildings located in a hot and humid climate. CFD simulations were used to evaluate airflow and thermal performance. The study demonstrated that appropriate natural ventilation strategies can improve indoor thermal conditions and reduce dependence on mechanical cooling. This research directly supports the CFD-based analysis proposed for the passive cooling house.

4. Li et al. (2024) – Natural Ventilation, Cool Roof and Shading

Li et al. experimentally and numerically investigated the combined effects of natural ventilation, external shading, cool roofs and thermal mass. The combined passive cooling approach reduced indoor air temperature by approximately **2.8°C** compared with the reference condition. The study concluded that combining several passive techniques provides better thermal performance than using a single technique. This supports the combination of ventilation, reflective surfaces and roof design in the proposed project.

5. Liu et al. (2023) – Cool Roof and Natural Ventilation

Liu et al. studied the optimization of residential top-floor rooms by combining cool roofs, natural ventilation and solar shading. The research showed that roof albedo, insulation thickness and window-to-wall ratio strongly influence building energy performance. The optimized passive design was able to achieve significant energy savings. This study supports the use of reflective roof surfaces and properly designed openings in the proposed house.

6. Sharma et al. (2021) – Thermal Comfort in Indian Naturally Ventilated Buildings

Sharma et al. reviewed thermal comfort research conducted in naturally ventilated buildings across the Indian subcontinent. The study found that occupants in Indian climates can have a broader thermal acceptance range than some international standards indicate. The research emphasizes the importance of adaptive thermal comfort when evaluating naturally ventilated buildings. This is particularly relevant to the proposed passive cooling house in Tamil Nadu.

7. Anand et al. (2023) – Passive Buildings

Anand et al. presented a state-of-the-art review of passive buildings and their energy-saving potential. The study emphasized the importance of building envelopes, including walls, roofs, doors, windows and openings, in controlling indoor thermal conditions. The findings support the use of appropriate envelope materials and passive architectural features in energy-efficient residential buildings.

8. Arumugam et al. (2022) – Insulation and Natural/Night Ventilation

Arumugam et al. reviewed passive techniques including insulation, phase change materials, natural ventilation and night ventilation under different climatic conditions. The study emphasized that passive techniques should be selected according to local climatic conditions. The findings support the use of insulation and natural ventilation together in hot-climate residential buildings.

9. Qin et al. (2022) – Experimental and Numerical Passive Cooling

Qin et al. investigated passive cooling using both experimental measurements and numerical simulations. The study focused on natural ventilation and indoor thermal environment and demonstrated the importance of opening location and airflow conditions. This supports the proposed design's use of strategically positioned openings for improving indoor airflow.

10. Nejat et al. (2021) – Windcatcher and Natural Ventilation

Nejat et al. experimentally and numerically investigated a windcatcher-based passive ventilation system. A small-scale model was tested experimentally and CFD simulations were validated against the experimental results. The research demonstrated that passive ventilation can provide adequate airflow and improve thermal comfort under suitable wind conditions. This supports the airflow-based design concept of the proposed project.

CHAPTER 3

COMPONENTS SPECIFICATION AND DESCRIPTION

3.1 COMPONENTS

The list of components used in Scaled Experimental Prototype are,

S.NO	COMPONENTS
1.	Cardboard
2.	Glue gun
3.	Foam board
4.	Temperature sensor (DS18B20IC)
5.	RTC module
6.	SD card module
7.	Memory card 8GB
8.	CR2032 Cell
9.	Jumper wire

3.1 List of Components in Scaled Experimental Prototype

3.2 COMPONENTS DESCRIPTION FULL SCALE APPLICATIONS

The list of components used in Full-Scale Residential Application

S.NO	MATERIAL
1.	Autoclaved Aerated Concrete (AAC) blocks
2.	Thin-bed AAC jointing mortar
3.	Reinforcement steel bars
4.	Cement–lime plaster
5.	Exterior solar-reflective paint
6.	Low-emissivity (Low-E) glass
7.	Steel door frames
8.	Natural clay tiles
9.	Insulated UPVC Door

Table 3.2 List of Components in Full-Scale Residential.

3.3 COMPONENTS DESCRIPTION

3.3.1 Carboard

Cardboard was used as the construction material for the experimental prototype due to its low cost, easy availability, and ease of fabrication. The cardboard model represents a scaled version of a residential structure to study passive cooling performance through natural ventilation and airflow-based cooling methods. Although cardboard does not replicate the exact thermal properties of real building materials, it is suitable for analyzing airflow patterns and indoor temperature variation. Cardboard has a relatively low thermal conductivity (approximately $0.05\text{--}0.10\text{ W/m}\cdot\text{K}$), low density, and low heat storage capacity, which help minimize external heat transfer and thermal lag effects. This allows clearer observation of ventilation-driven cooling performance. The cardboard prototype serves as a proof-of-concept model for validating passive cooling strategies before full-scale implementation.



Figure 3.3.1 – Carboard

3.3.2 Glue Gun

A hot glue gun was used as the primary bonding method for assembling the cardboard prototype due to its quick setting time, ease of application, and sufficient bonding strength for lightweight materials. Hot melt adhesive provides adequate mechanical stability while allowing rapid fabrication and minor design modifications during experimentation. The adhesive exhibits low thermal conductivity (approximately $0.15\text{--}0.25\text{ W/m}\cdot\text{K}$), which minimizes additional heat transfer at joints and prevents localized thermal bridging. Since the glue is applied in thin layers, its influence on overall thermal behavior is negligible. The use of a glue gun ensures structural integrity of the prototype without significantly affecting airflow patterns or thermal performance, making it suitable for experimental validation of passive cooling strategies.



Figure 3.3.2 Glue Gun

3.3.3 Foam board

Foam board was used as a supplementary construction material in the experimental prototype due to its lightweight nature, ease of cutting, and good thermal insulation properties. The foam board model represents wall and partition elements of a scaled residential structure to support the study of passive cooling through natural ventilation. Foam board has a very low thermal conductivity (approximately $0.03\text{--}0.05\text{ W/m}\cdot\text{K}$), low density, and low heat storage capacity, which significantly reduce conductive heat transfer from the surroundings. These properties help maintain stable indoor conditions, allowing clearer observation of airflow-driven cooling effects. The use of foam board enhances insulation performance while remaining suitable for rapid prototyping and proof-of-concept validation.



Figure 3.3.3 Foam board

3.3.4 Temperature sensor (DS18B20IC)

The DS18B20 temperature sensor was used to measure indoor and outdoor temperatures of the experimental prototype with high accuracy and reliability. The sensor provides digital temperature output using a 1-Wire communication protocol, reducing wiring complexity and improving data stability. It has a measurement range of $-55\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$ with an accuracy of $\pm 0.5\text{ }^{\circ}\text{C}$ in the operating range relevant to building thermal studies. The small size and fast response time of the DS18B20 make it suitable for real-time monitoring of temperature variation caused by passive cooling and natural ventilation. The sensor enables precise comparison of thermal performance under different experimental conditions.



Figure 3.3.4 Temperature sensor (DS18B20IC)

3.3.5 RTC Module

A Real-Time Clock (RTC) module was used to accurately record the time and date associated with temperature measurements during the experimental study. The RTC enables precise time-stamping of sensor data, which is essential for analysing temperature variation with respect to hourly and diurnal changes. It operates independently of the main power supply using a backup battery, ensuring continuous timekeeping during power interruptions. The RTC module improves data reliability and repeatability by synchronizing temperature readings with real-time environmental conditions. Its integration allows systematic data logging and supports detailed thermal performance analysis of the passive cooling prototype.

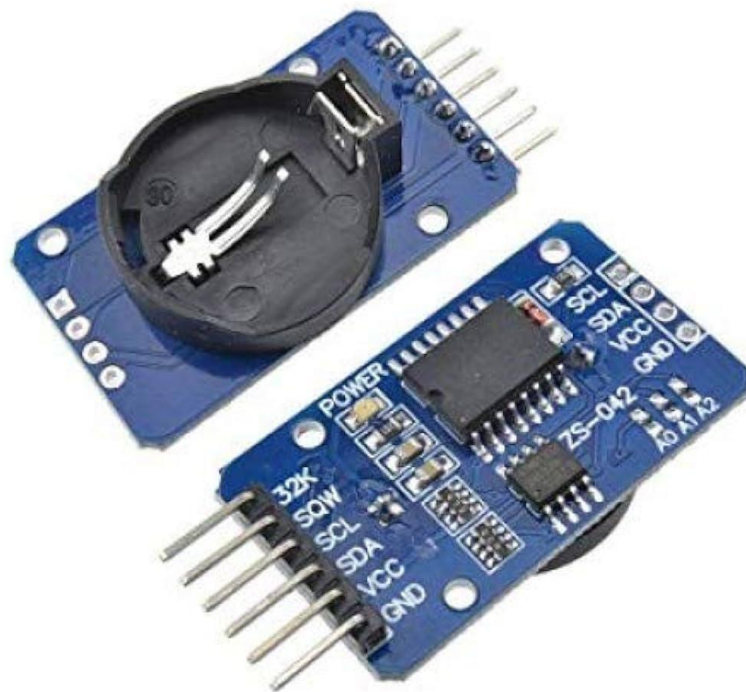


Figure 3.3.5 RTC Module

3.3.6 SD Card Module

An SD card module was used for data logging and long-term storage of temperature and time-stamped measurements obtained during the experimental study. The module enables real-time recording of sensor data, allowing continuous monitoring of indoor and outdoor thermal conditions. It communicates with the controller using the SPI protocol, ensuring reliable and high-speed data transfer. The SD card provides non-volatile storage, preventing data loss during power interruptions. The recorded data can be easily transferred to a computer for post-processing, graphical analysis, and comparison of passive cooling performance under different environmental conditions.



Figure 3.3.6 SD Card Module

3.3.7 Memory card 8GB

An 8 GB memory card was used as the primary data storage medium for recording temperature and time-stamped measurements obtained during the passive cooling experiment. The storage capacity of 8 GB is sufficient to store large volumes of data collected at short sampling intervals over extended monitoring periods without the risk of memory overflow. The memory card provides non-volatile storage, ensuring that recorded data remains محفوظ even during unexpected power interruptions or system shutdowns. It supports reliable read and write operations when interfaced with the SD card module and microcontroller. The stored data can be easily transferred to a computer for post-processing, graphical representation, and detailed thermal performance analysis. The use of an 8 GB memory card enables continuous data logging, improves data integrity, and supports accurate comparison of indoor and outdoor temperature variations under different environmental and experimental conditions, thereby enhancing the reliability of the overall passive cooling study.



Figure 3.3.7 Memory Card 8 GB

3.3.8 CR2032 Cell

A CR2032 coin cell battery was used as the backup power source for the Real-Time Clock (RTC) module in the experimental setup. The CR2032 provides a nominal voltage of 3 V and offers stable, long-lasting power consumption suitable for low-current electronic components. Its primary function is to maintain accurate time and date information when the main power supply is turned off or interrupted. This ensures continuity in time-stamped temperature data and prevents loss of chronological accuracy during data logging. The compact size and reliable discharge characteristics of the CR2032 make it ideal for embedded applications requiring continuous timekeeping. The coin cell has a low self-discharge rate, allowing long operational life without frequent replacement. The use of a CR2032 battery enhances the reliability of data acquisition, supports uninterrupted monitoring, and improves the overall accuracy and integrity of the passive cooling performance evaluation.



Figure 3.3.8 CR2032 cell

3.3.9 Autoclaved Aerated Concrete (AAC) blocks

Autoclaved Aerated Concrete (AAC) blocks were considered as a reference construction material for evaluating the applicability of the proposed passive cooling strategies in real residential buildings. AAC blocks are lightweight, porous, and possess excellent thermal insulation properties due to the presence of uniformly distributed air voids. The thermal conductivity of AAC blocks typically ranges between 0.10 and 0.16 W/m·K, which is significantly lower than that of conventional clay bricks and concrete. This property reduces conductive heat transfer through building walls and helps maintain lower indoor temperatures. AAC blocks also exhibit low density and moderate thermal mass, contributing to reduced heat storage and delayed heat gain. In addition, they provide adequate structural strength, fire resistance, and sound insulation, making them suitable for sustainable construction. The inclusion of AAC blocks in this study supports the scalability of the experimental results and demonstrates the feasibility of implementing the proposed passive cooling techniques in full-scale residential buildings.



Figure 3.3.9 AAC Block

3.3.10 Thin-bed AAC Jointing Mortar

Thin-bed AAC jointing mortar was used to assemble Autoclaved Aerated Concrete (AAC) blocks in the prototype and real-scale building simulations. This specialized mortar is designed for thin-layer application, typically 1–3 mm, ensuring minimal thermal bridging between blocks while providing adequate structural bonding. It has low thermal conductivity (approximately 0.30–0.50 W/m·K), which reduces heat transfer through the wall joints and maintains the overall insulation performance of AAC walls. The mortar sets quickly, allows precise alignment of blocks, and supports lightweight construction. Its use ensures that the thermal and structural properties of the AAC walls are preserved, enabling accurate evaluation of passive cooling strategies. By minimizing joint thickness and enhancing bonding, thin-bed AAC mortar contributes to efficient heat resistance, stable indoor temperatures, and improved performance of natural ventilation and passive cooling techniques in full-scale buildings.



Figure 3.3.10 Thin-bed AAC Jointing Mortar

3.3.11 Reinforcement Steel Bar

Reinforcement steel bars were used to provide structural stability and load-bearing capacity in the building system incorporating Autoclaved Aerated Concrete (AAC) blocks. AAC blocks possess good compressive strength but require reinforcement to resist tensile stresses, bending, and shear forces, particularly in lintels, beams, columns, and roof elements. Steel reinforcement ensures overall structural integrity and durability of the construction. Although steel has high thermal conductivity (approximately 45–60 W/m·K), its usage is limited to critical structural zones to minimize thermal bridging effects. Proper placement and adequate concrete cover reduce direct heat transfer through reinforced elements. The inclusion of reinforcement steel bars ensures safety, stability, and compliance with structural design requirements, while maintaining compatibility with passive cooling strategies. This allows the proposed cooling concepts to be implemented in real-scale residential buildings without compromising structural performance or thermal comfort.



Figure 3.3.11 Reinforcement Steel Bar

3.3.12 Cement–lime Plaster

Cement–lime plaster was used as the surface finishing material for walls constructed with Autoclaved Aerated Concrete (AAC) blocks. The addition of lime to cement plaster improves workability, adhesion, and surface smoothness while reducing shrinkage cracks. Cement–lime plaster exhibits moderate thermal conductivity (approximately $0.60\text{--}0.90\text{ W/m}\cdot\text{K}$), which provides a balanced combination of surface protection and thermal resistance. The plaster layer acts as a protective barrier against moisture ingress, weathering, and surface deterioration, thereby enhancing the durability of the wall system. Lime content also improves vapor permeability, allowing walls to “breathe” and reducing internal moisture accumulation, which contributes to improved indoor thermal comfort. When applied in controlled thickness, cement–lime plaster does not significantly increase heat gain and supports the overall performance of passive cooling strategies. Its use ensures long-term structural protection while maintaining acceptable thermal and environmental performance in residential buildings.



Figure 3.3.12 Cement – lime Plaster

3.3.13 Exterior Solar-reflective Paint

Exterior solar-reflective paint was applied to the outer surfaces of the building envelope to reduce solar heat gain and improve thermal performance. This type of coating contains reflective pigments that reflect a significant portion of incoming solar radiation, particularly in the infrared range. As a result, the surface temperature of walls and roofs is reduced compared to conventional paints. Solar-reflective paints typically exhibit high solar reflectance and low thermal emissivity, which help limit heat absorption during peak sunlight hours. By lowering external surface temperatures, the heat transfer into the indoor space through conduction is reduced, contributing to improved indoor thermal comfort. The use of solar-reflective paint enhances the effectiveness of passive cooling strategies, especially in hot climatic conditions. It is a cost-effective, easily applicable solution that supports energy-efficient building design and reduces dependence on active cooling systems.



Figure 3.3.13 External Solar-reflective Paint

3.3.14 Low-emissivity (Low-E) Glass

Low-emissivity (Low-E) glass was considered for window openings to reduce solar heat gain while maintaining adequate natural daylight. Low-E glass is coated with a thin, transparent metallic layer that reflects long-wave infrared radiation and limits heat transfer through glazing. This coating significantly reduces the emissivity of the glass surface, thereby minimizing radiative heat exchange between the outdoor environment and indoor space. Compared to conventional clear glass, Low-E glass offers lower thermal transmittance (U-value) and reduced solar heat gain coefficient (SHGC), which helps maintain lower indoor temperatures under high solar exposure. The use of Low-E glass supports passive cooling by decreasing unwanted heat entry through windows, which are typically major sources of thermal gain in buildings. At the same time, it allows sufficient visible light transmission, reducing the need for artificial lighting and improving overall energy efficiency and indoor thermal comfort.



Figure 3.3.14 Low-emissivity Glass

3.3.15 Natural Clay Tiles

Natural clay tiles were considered as a roofing material due to their favorable thermal and environmental properties. Clay tiles have relatively low thermal conductivity (approximately 0.6–1.0 W/m·K) and high thermal mass, which helps delay heat transfer into the indoor space during peak solar exposure. The inherent porosity of clay allows limited air circulation within the tile structure, aiding in heat dissipation. When used with an air gap beneath the roof, clay tiles further reduce heat gain by promoting convective cooling. Their natural color and surface texture also reflect a portion of solar radiation, lowering roof surface temperatures compared to metal roofing. In addition to thermal benefits, clay tiles are durable, fire-resistant, and environmentally sustainable. The use of natural clay tiles enhances roof insulation performance and complements passive cooling strategies by reducing indoor temperature rise and improving thermal comfort in residential buildings.



Figure 3.3.15 Natural Clay Tiles

3.3.16 Insulated UPVC Door

An insulated uPVC (unplasticized polyvinyl chloride) door was considered for the building envelope to reduce heat transfer while maintaining durability and airtightness. uPVC doors have low thermal conductivity (approximately $0.16\text{--}0.20\text{ W/m}\cdot\text{K}$) and are often constructed with multi-chamber profiles and insulating cores, which significantly reduce conductive heat gain. Compared to conventional wooden or metal doors, insulated uPVC doors provide better thermal resistance and minimize air leakage. This helps maintain stable indoor temperatures and enhances the effectiveness of passive cooling strategies. The door also offers good resistance to moisture, corrosion, and weathering, making it suitable for hot and humid climatic conditions. Additionally, uPVC doors require minimal maintenance and provide adequate acoustic insulation. The use of an insulated uPVC door contributes to improved building envelope performance, reduced unwanted heat entry, and enhanced indoor thermal comfort in energy-efficient residential buildings.



Figure 3.3.16 Insulated UPVC Door

CHAPTER 4

WORKING PRINCIPLE

WORKING PRINCIPLE OF THE PROPOSED PASSIVE COOLING HOME

4.1. Objective of the Proposed System

The primary objective of the present study is to reduce indoor temperature using only passive cooling techniques, without employing any mechanical or electrical cooling systems. The proposed system aims to achieve thermal comfort through architectural form, natural airflow control, and heat rejection mechanisms, making it suitable for energy-efficient and sustainable building applications in hot climatic regions.

4.2. Description of the Developed Prototype

The present work focuses on a semi-cylindrical dome roof-based passive cooling prototype, developed using cardboard material for concept validation. The prototype represents a scaled physical model intended to study airflow behaviour and thermal performance rather than material strength.

The semi-cylindrical dome roof is selected due to its corner-free geometry, which reduces heat accumulation zones and promotes smoother air circulation. Unlike conventional flat or angular roofs, the curved dome surface distributes incident solar radiation more uniformly, thereby reducing localized heat gain. Additionally, the absence of sharp corners inside the structure allows warm air to move upward freely without obstruction.

A small opening is provided at the top of the dome, which acts as a natural exit point for hot air. Large window openings are provided on the side walls to facilitate enhanced air entry and Venturi-assisted airflow. A long roof overhang is incorporated to minimize direct solar radiation on the vertical walls, further reducing indoor heat gain.

4.3 Passive Cooling Methods Adopted

The cooling mechanism of the proposed system is based entirely on natural physical phenomena, primarily:

- Stack effect (buoyancy-driven ventilation)
- Venturi effect (pressure-driven airflow acceleration)

These passive methods work together to establish a continuous airflow cycle that removes warm indoor air and replaces it with cooler ambient air.

4.4. Heat Gain Reduction Mechanism

Heat gain inside the prototype occurs mainly due to solar radiation on the roof and walls. To minimize this effect, a reflective surface coating is applied to the exterior, which reduces solar heat absorption. The heat gained due to solar radiation can be expressed as:

$$Q_{\text{solar}} = \alpha I A$$

By reducing surface absorptivity, the amount of heat entering the indoor space is significantly lowered. The semi-cylindrical roof further reduces effective heat gain by distributing radiation across a curved surface instead of concentrating it on flat planes.

4.5. Airflow Path in the Proposed Prototype

The airflow path in the developed prototype is intentionally guided by structural geometry. Ambient air enters the indoor space through large window openings located on one side of the structure, aligned with the prevailing airflow direction. This incoming air flows across the occupied region and exits through openings on the opposite side of the wall.

As the air absorbs sensible heat from the indoor environment, its temperature increases, causing a reduction in air density. The warmed air naturally rises and moves upward along the curved inner surface of the semi-cylindrical dome. Due to the absence of corners, airflow experiences minimal resistance and turbulence during upward movement.

At the highest point of the dome, a top opening function as a passive air extractor, allowing accumulated hot air to escape naturally.

4.6. Stack Effect in the Semi-Cylindrical Dome

The stack effect plays a dominant role in removing hot air from the prototype. Since warm air is lighter than cool air, it rises toward the dome apex, creating a pressure difference between the lower and upper regions. The pressure difference generated by buoyancy can be expressed as:

$$\Delta P = \rho g H \left(\frac{T_i - T_o}{T_o} \right)$$

This pressure difference continuously drives warm air out through the top opening, while cooler air is drawn inside from the side openings, maintaining natural ventilation.

4.7. Venturi Effect and Enhanced Air Circulation

To further improve airflow, the proposed prototype utilizes the Venturi effect through large window openings and comparatively smaller exit regions. When air flows from a larger cross-sectional area to a smaller area, its velocity increases, resulting in a reduction in static pressure.

The continuity equation governing this behaviour is:

$$A_1V_1=A_2V_2$$

According to Bernoulli's principle:

$$P + \frac{1}{2}\rho V^2 = \text{constant}$$

The reduction in pressure near the outlet assists in drawing air through the structure, strengthening indoor air circulation. This Venturi-assisted airflow improves heat removal and enhances thermal comfort without any mechanical support.

4.8. Role of the Top Air Extractor

The top opening provided at the apex of the semi-cylindrical dome acts as a passive air extractor. This feature is critical to the performance of the proposed system, as it prevents heat accumulation near the roof and ensures continuous evacuation of warm air.

Because the dome has no corners, warm air does not get trapped and can exit smoothly through the extractor. This design significantly improves natural airflow efficiency compared to conventional roof geometries.

4.9. Sensible Heat Removal by Airflow

The removal of sensible heat from the indoor space occurs due to continuous air exchange. The rate of heat removal can be expressed as:

$$Q_{air} = \dot{m}c_p(T_i - T_o)$$

4.10 Experimental Observation and Performance Validation

During the experimental evaluation of the prototype, indoor and outdoor temperatures were monitored, and airflow behaviour was qualitatively observed. The indoor space consistently exhibited lower temperatures compared to outdoor conditions, particularly during peak daytime periods.

Additionally, noticeable airflow movement was felt inside the structure, confirming the effectiveness of the combined stack and Venturi effects. These observations validate the conceptual effectiveness of the proposed design methodology.

4.11 Conclusion of the Working Principle

The working principle of the proposed passive cooling prototype confirms that indoor temperature reduction can be achieved without the use of mechanical or electrical systems. The semi-cylindrical dome roof plays a key role by minimizing heat accumulation due to its corner-free geometry and by enabling smooth upward movement of warm air. The curved roof surface supports uniform heat distribution and prevents thermal stagnation inside the structure. A top opening provided at the dome apex functions as a natural air extractor, allowing hot air to escape due to buoyancy forces, thereby strengthening the stack effect. Large window openings improve airflow circulation through the Venturi effect, increasing air velocity and enhancing heat removal from the indoor space. Additionally, the use of reflective surface treatment and extended roof overhangs reduces direct solar heat gain on walls. Observations of indoor and outdoor temperatures indicate effective cooling performance, while noticeable airflow movement validates the effectiveness of the integrated passive cooling approach.

CHAPTER 5

COST ESTIMATION

5.1 MINIMUM COST ESTIMATION FOR REAL APPLICATION

S.NO	NAME OF THE COMPONENT	QUANTITY	PRICE(Rs.)
1.	Autoclaved Aerated Concrete (AAC) blocks	As per requirement	18000
2.	Thin-bed AAC jointing mortar	As per requirement	3000
3.	Reinforcement steel bars	As per requirement	6500
4.	Cement–lime plaster	As per requirement	4200
5.	Exterior solar-reflective paint	As per requirement	2800
6.	Low-emissivity (Low-E) glass	As per requirement	5500
7.	Steel door frames	1 set	2200
8.	Natural clay tiles	As per requirement	7000
9.	Insulated UPVC Door	1	6800
10.	Total		56000

Table 5.1 Cost Estimation

5.2 PROTOTYPE COST ESTIMATION

S.NO	NAME OF THE COMPONENT	QUANTITY	PRICE (Rs.)
1.	Cardboard	Small sheets	150
2.	Glue gun (with glue sticks)	1	200
3.	Foam board	Small sheets	150
4.	Temperature sensor (DS18B20 IC)	2	240
5.	RTC module	1	150
6.	SD card module	1	150
7.	Memory card (8 GB)	1	250
8.	CR2032 Cell	1	50
9.	Jumper wires	1 set	80
10.	Total		1420

Table 5.2 Prototype Cost Estimations

CHAPTER 6

PROTOTYPE

A physical prototype was developed to demonstrate and validate the proposed passive cooling concept under controlled conditions. The prototype represents a simplified model of a naturally ventilated structure incorporating a semi-cylindrical dome roof and passive airflow features. Cardboard material was selected for prototype fabrication due to its low cost, ease of handling, and suitability for rapid design iteration. The objective of the prototype is concept validation rather than structural performance evaluation.

The prototype consists of a semi-cylindrical dome roof mounted over enclosed side walls, forming a continuous curved surface without sharp corners. This geometry facilitates smooth airflow movement and minimizes heat accumulation within the enclosed space. Large window openings are provided on opposite sides of the structure to allow ambient air entry and exit, supporting cross ventilation and Venturi-assisted airflow. A small opening is positioned at the top of the dome, which acts as a passive air extractor for the removal of hot air.

6.1 3D CAD MODELING

6.1.1 Isometric View

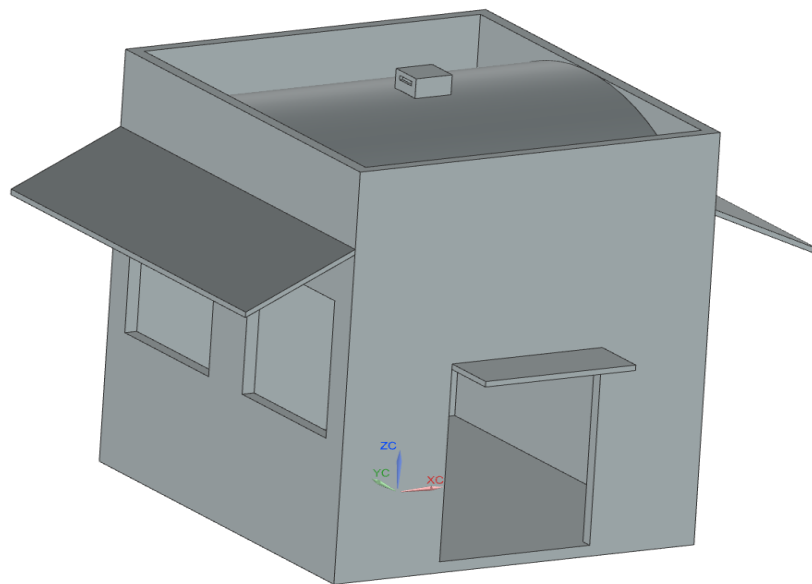


Figure 6.1.1 Isometric View

6.1.2 Right Side View

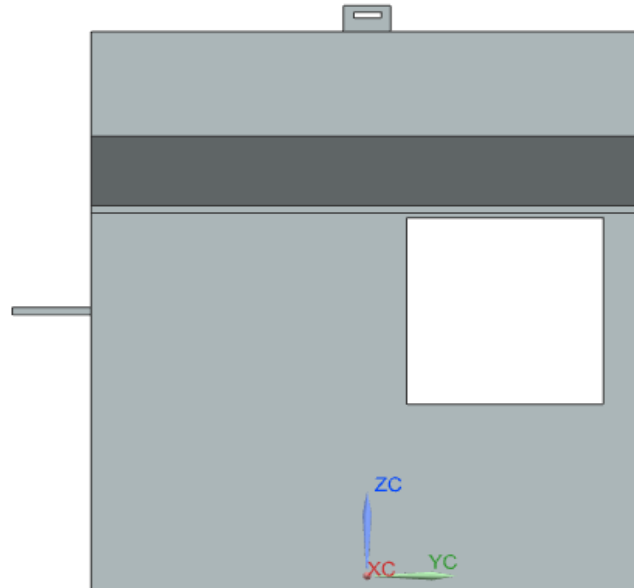


Figure 6.1.2 Right Side View

6.1.3 Left Side View

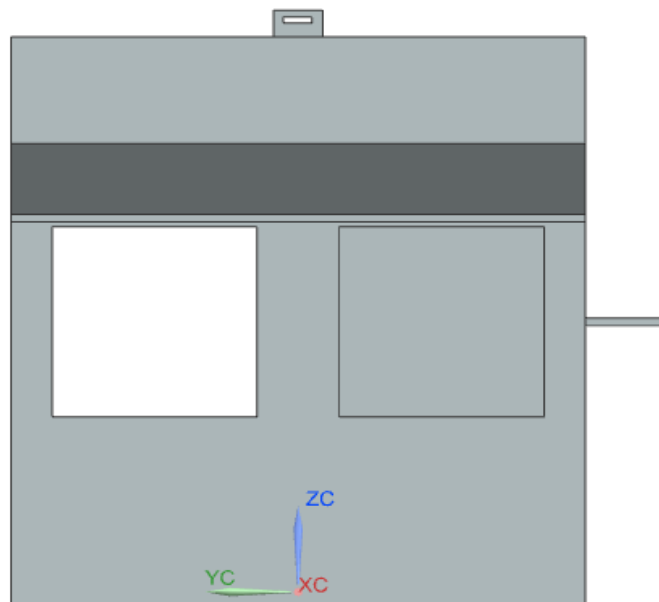


Figure 6.1.3 Left Side View

6.1.4 Top View

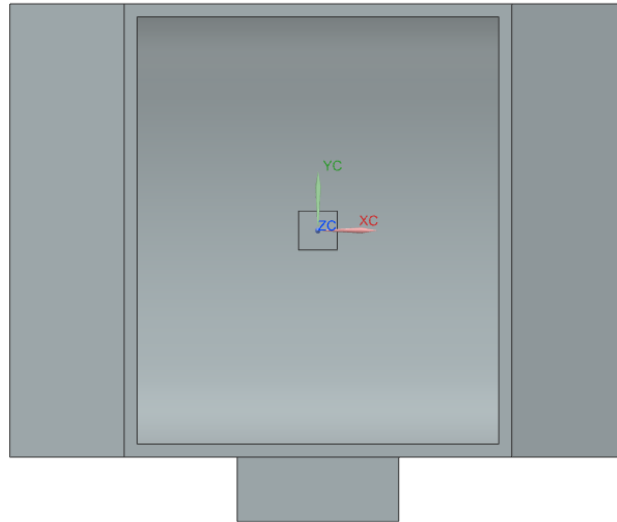


Figure 6.1.4 Top View

6.1.5 Front View

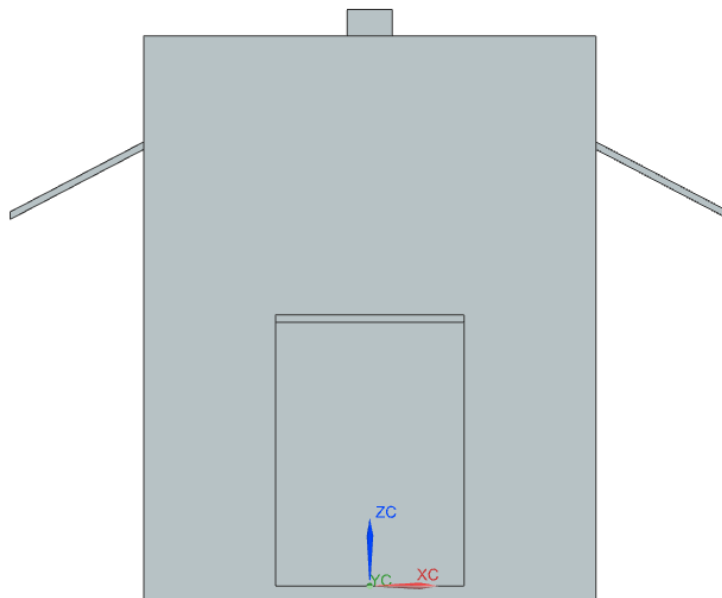


Figure 6.1.5 Front View

6.2 PROTOTYPE VIEWS

6.2.1 Front View



Figure 6.2.1 Front View

6.2.2 Side View



Figure 6.2.2 Side View

6.2.3 Top View



Figure 6.2.3 Top View

6.2.4 Semi Cylindrical Dome View



Figure 6.2.4 Semi Cylindrical Dome View

CHAPTER 7

RESULTS AND DISCUSSION

The experimental evaluation of the developed passive cooling prototype was carried out under real outdoor exposure conditions. The measured output values (t_1 and t_2) were recorded at regular intervals and documented in an Excel sheet for systematic analysis. Here, t_1 represents the external condition, while t_2 represents the internal condition within the prototype.

The recorded results show a consistent and noticeable difference between t_1 and t_2 throughout the observation period. While t_1 exhibited a gradual increase, the corresponding values of t_2 remained comparatively lower and showed a slower rate of rise. This behaviour indicates that the proposed design effectively reduces heat penetration into the enclosed space.

The maximum observed difference between t_1 and t_2 was approximately 2–3 units, particularly during peak external exposure. This reduction demonstrates the effectiveness of the semi-cylindrical dome structure combined with surface reflection and natural air circulation. The curved geometry minimizes direct exposure, while the internal air movement delays heat transfer.

The stable performance of t_2 , even under increasing external conditions, confirms that the passive cooling strategy provides improved thermal comfort without the use of mechanical or electrical systems. Overall, the results validate the feasibility and efficiency of the proposed passive cooling prototype for sustainable building applications.

In addition, the observed performance highlights the reliability of the prototype under varying external exposure conditions. The gradual variation in t_2 compared to t_1 indicates reduced heat fluctuation inside the structure. This stability suggests improved thermal insulation behaviour achieved through the dome geometry and material selection. The results also confirm that passive design strategies can effectively control internal conditions without active energy input. Hence, the proposed model demonstrates strong potential for low-energy and climate-responsive building design.

time	t1	t2
9.30	30.12	27.55
10.00	29.1	27.01
10.30	29.3	26.8
11.00	28.9	26.3
11.30	29.6	26.89
12.00	28.5	26.19
12.30	27.25	25.4
1.00	26.9	25.17

Figure 7.1 Excel Data Output

CHAPTER 8

SCOPE OF THE FUTURE WORK

- ✓ The prototype can be developed using real construction materials such as AAC blocks, hollow clay blocks, or fly ash bricks to evaluate performance in permanent buildings.
- ✓ Different dome and curved roof geometries can be analysed to identify the most effective shape for reducing heat gain.
- ✓ Advanced passive techniques like wind catchers, stack ventilation, roof ventilators, and earth air tunnels can be integrated to improve airflow.
- ✓ Computational simulations using ANSYS or Energy Plus can be performed to validate experimental results.
- ✓ Long-term monitoring can be carried out to study performance under different seasonal conditions.
- ✓ Smart sensors and IoT-based data logging systems can be added for continuous monitoring and analysis.
- ✓ The design can be scaled for full-size residential houses and low-cost housing projects.
- ✓ Economic analysis and life-cycle assessment can be conducted to evaluate cost effectiveness and sustainability.
- ✓ Renewable energy systems such as solar panels can be combined with the passive cooling design.
- ✓ Design guidelines can be developed for different climatic regions.

8.1 ADVATAGES:

- ✓ Reduces indoor heat gain without using mechanical or electrical systems.
- ✓ Low operating cost and energy-efficient solution.
- ✓ Environmentally friendly and sustainable design approach.
- ✓ Simple construction with locally available materials.
- ✓ Requires minimal maintenance compared to active cooling systems.
- ✓ Suitable for hot and tropical climatic conditions.
- ✓ Improves thermal comfort naturally.
- ✓ Reduces dependency on air conditioners and power consumption.

8.2 APPLICATIONS:

- ✓ Residential buildings in hot and dry regions.
- ✓ Low-cost and affordable housing projects.
- ✓ Rural housing and village development schemes.
- ✓ Temporary shelters and emergency housing.
- ✓ Educational buildings such as classrooms and hostels.
- ✓ Community halls and public utility buildings.
- ✓ Eco-friendly and green building projects.
- ✓ Sustainable architecture and research demonstrations

CHAPTER 9

APPENDICES

APPENDIX – 1 FIELD VISIT

As part of our project study, we visited a naturally cooled residential house located near Dindigul to understand how passive cooling works in real buildings. The house was designed to reduce heat inside the living space by using natural methods instead of electrical cooling systems.

The building uses thick walls, light-coloured exterior surfaces, and a roof design that helps reduce direct heat entry. Openings such as windows and vents are placed in a way that allows fresh air to move easily through the house. Shaded areas and extended roof edges were also observed, which help block direct sunlight during peak daytime conditions.

The indoor environment of the house felt more comfortable compared to the outdoor surroundings, showing that passive design methods are effective in hot regions. This visit helped us clearly understand how passive cooling principles are applied in real houses and helped us improve our project design based on practical construction methods.



Figure 9.1 Field Visit

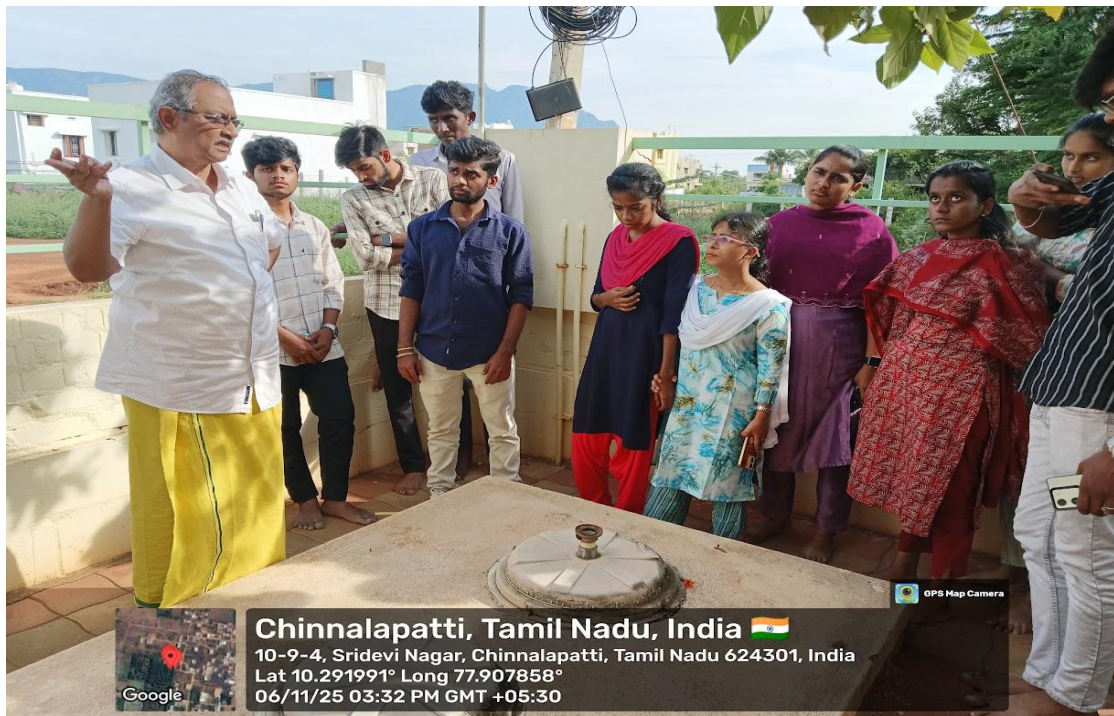


Figure 9.2 Discussion on Passive Cooling Techniques



Figure 9.3 Solar Panel Inspection

APPENDIX – 2

COMPONENTS SPECIFICATION

PROTOTYPE SPECIFICATION

1. Cardboard

Type: corrugated cardboard

Thickness: 5 mm

Properties: Lightweight, low cost, easy to cut and shape

Application: Structural framework of the prototype

2. Glue Gun

Type: Hot glue gun

Operating temperature: 170–190 °C

Adhesive type: Hot melt glue sticks

Applications: joining and fixing prototype components

3. Foam Board

Thickness: 5mm

Material: Polystyrene foam with paper coating

Properties: Lightweight, moderate thermal insulation

Application: Walls and roof surface of the prototype

4. Temperature Sensor (DS18B20 IC)

Sensor Type: Digital temperature sensor

Temperature Range: –55 °C to +125 °C

Accuracy: ± 0.5 °C

Output: Digital (1-Wire interface)

5. RTC Module

Module Type: DS3231

Operating Voltage: 3–5 V

Function: Real-time time and date tracking

Backup: Coin cell supported

6. SD Card Module

Interface: SPI communication

Operating Voltage: 5 V

Function: Data logging and storage

7. Memory Card

Type: Micro SD card

Storage Capacity: 8 GB

Application: Storing temperature and time data

8. CR2032 Cell

Type: Lithium coin cell

Voltage Rating: 3 V

Application: RTC module backup power

9. Jumper Wire

Type: Female –Female

Length: 20–30 cm

Application: Electrical connections between modules

REAL SCALE BUILDING SPECIFICATION

1. Autoclaved Aerated Concrete (AAC) Blocks

Density: 550–650 kg/m³

Size: Standard AAC block size

Properties: Lightweight, good thermal insulation, fire resistant

Application: Wall construction

2. Thin-bed AAC Jointing Mortar

Type: Factory-prepared thin-bed mortar

Joint Thickness: 3–5 mm

Properties: High bonding strength, low shrinkage

Application: Bonding AAC blocks

3. Reinforcement Steel Bars

Material: Mild steel

Diameter: As per structural requirement

Properties: High tensile strength

Application: Structural reinforcement

4. Cement–Lime Plaster

Composition: Cement, lime, and sand

Thickness: 12–15 mm

Properties: Smooth finish, improved durability

Application: Wall plastering

5. Exterior Solar-Reflective Paint

Type: Heat-reflective exterior coating

Reflectance: High solar reflectance index

Properties: Reduces heat absorption

Application: External wall and roof surfaces

6. Low-Emissivity (Low-E) Glass

Glass Type: Coated energy-efficient glass

Thickness: 5–6 mm

Properties: Reduces heat transfer, allows daylight

Application: Windows and openings

7. Steel Door Frames

Material: Mild steel

Thickness: As per standard

Properties: Strong, durable, corrosion resistant

Application: Door framing

8. Natural Clay Tiles

Material: Baked natural clay

Properties: Thermal insulation, weather resistant

Application: Roofing

9. Insulated UPVC Door

Material: UPVC with insulation core

Properties: Low heat transfer, moisture resistant

Application: Main entrance door

APPENDIX – 3

PSEUDO CODE FOR KEY FUNCTIONS

This document presents structured pseudocode for the data acquisition system used in the passive cooling experiment. The system reads temperature data from two sensors and stores the readings on an SD card for later analysis.

1. System Initialization

BEGIN System_Initialization

 Initialize Temperature_Sensor_1

 Initialize Temperature_Sensor_2

 Initialize SD_Card_Module

 IF SD_Card_Module NOT detected THEN

 Display "Error: SD Card not found"

 TERMINATE program

 END IF

 Create or Open file 'temperature_log.txt' in SD Card

 Display "System Ready"

END System_Initialization

2. Temperature Data Acquisition

FUNCTION Read_Temperature_Sensors

 Temp1 $\leftarrow$ Read value from Temperature_Sensor_1

 Temp2 $\leftarrow$ Read value from Temperature_Sensor_2

 RETURN Temp1, Temp2

END FUNCTION

3. Temperature Difference Calculation

FUNCTION Calculate_Temperature_Difference(Temp1, Temp2)

 Temp_Difference $\leftarrow$ Temp1 – Temp2

 RETURN Temp_Difference

END FUNCTION

4. Data Logging to SD Card

FUNCTION Store_Data_To_SD(Time, Temp1, Temp2, Temp_Difference)

Open 'temperature_log.txt'

Write Time, Temp1, Temp2, Temp_Difference to file

Close file

END FUNCTION

5. Main Control Loop

BEGIN Main_Program

Call System_Initialization

WHILE System_Power = ON DO

(Temp1, Temp2) ← Read_Temperature_Sensors()

Temp_Difference ← Calculate_Temperature_Difference(Temp1, Temp2)

Time ← Get_Current_Time()

Call Store_Data_To_SD(Time, Temp1, Temp2, Temp_Difference)

Wait for Sampling_Interval (e.g., 5 seconds)

END WHILE

END Main_Program

6. System Shutdown

BEGIN System_Shutdown

Ensure all files are closed

Stop Temperature Sensors

Display "System Stopped"

END System_Shutdown

CHAPTER 10

CONCLUSION

This project successfully demonstrated the concept and effectiveness of a passive cooling system through the design and development of a small-scale prototype. The main objective was to reduce indoor temperature without using any mechanical or electrical cooling devices, relying only on natural principles such as airflow, heat dissipation, and suitable structural geometry. A semi-cylindrical dome-shaped roof structure was selected because of its ability to reduce heat gain, enhance air circulation, and minimize direct solar radiation compared to conventional flat roofs. The prototype was constructed using cardboard, foam board, and basic joining materials to represent the real-world structure in a simplified and economical manner. Although cardboard is not a permanent construction material, it was chosen for experimental validation due to its low cost, ease of fabrication, and suitability for testing airflow and thermal behaviour at a model level. Temperature sensors were used to observe indoor and outdoor temperature variations, which helped in understanding the cooling performance of the design. The results obtained from the prototype indicate that the passive cooling approach can effectively lower indoor temperature when compared to the surrounding outdoor environment, especially during peak daytime hours. Natural ventilation through openings and the dome profile played a significant role in maintaining thermal comfort. This confirms that architectural design, orientation, and material selection are critical factors in reducing heat accumulation inside buildings. Overall, this project proves that passive cooling techniques can be a sustainable, energy-efficient, and eco-friendly alternative to conventional cooling systems. The study highlights the potential of applying such designs in real residential buildings using appropriate construction materials like AAC blocks, insulation layers, and reflective coatings. Future work can include full-scale simulations, advanced material analysis, and long-term thermal performance studies to further enhance the effectiveness of the proposed system.

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